

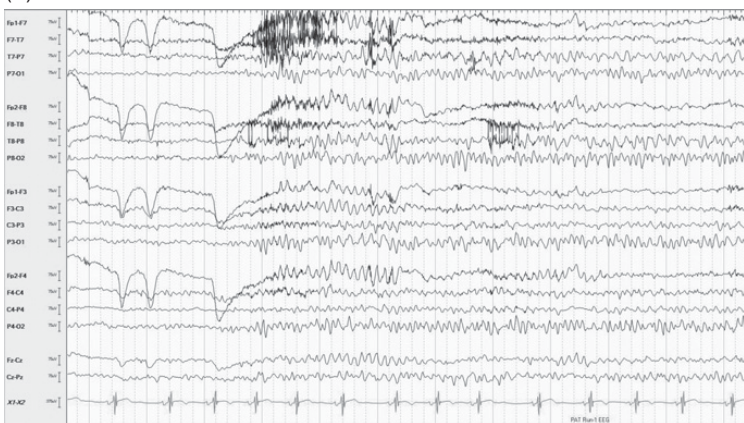
The specific arrangement of channels (electrode pairs) on an EEG display is called a *montage* (French for “assembly”). This arrangement of channels allows the reader to analyze a waveform from a specific point of view. The reader may conceptualize montages to resemble the cuts of a head scan (such as axial, sagittal, or coronal) as each montage observes the same electrographic activity from a specific angle. Multiple designs exist but overall, montages can be grouped into one of two basic types:

1. bipolar
2. referential

Bipolar montages record the difference in potential between two adjacent electrodes. Both these electrodes intend to measure cortical electrical activity, that is, both are active electrodes. Bipolar montages are named based on the plane their electrode pairs are arranged. Longitudinal bipolar montage (also called a double banana based on its resemblance) and transverse bipolar montages are the two most commonly used bipolar montages [1]. As described earlier, most of the EEGs in this book use this montage. Other types of bipolar montages include the coronal and circumferential montages. Their advantages are explained later in Chapter 4. Figure 3.1 shows the display (A) and diagram (B) of electrode pairs in a longitudinal bipolar montage while Figure 3.2 shows the display (A) and diagram (B) of electrode pairs in a transverse montage.

Referential montages record the difference in potential between two distant electrodes. One of the two electrodes is active (records cerebral activity) while the other, relatively inactive electrode, is intended to serve as a “reference.” The ipsilateral, contralateral ear or vertex are commonly used points of references as these are usually neutral. The computed average potential of all the active electrodes may also be used as a hypothetical reference point (average reference) [1]. Figure 3.3 shows the display (A) and diagram (B) of electrode pairs in an ipsilateral ear reference montage while Figure 3.4 shows the display of an average reference montage.

(A)



(B)

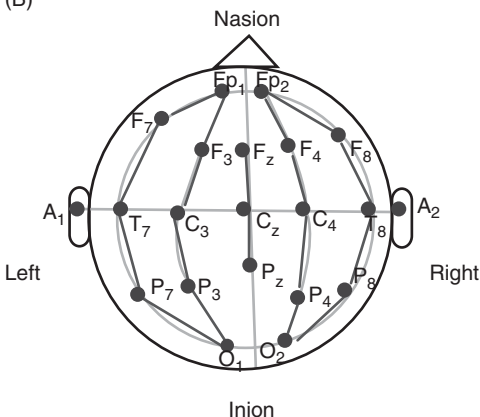
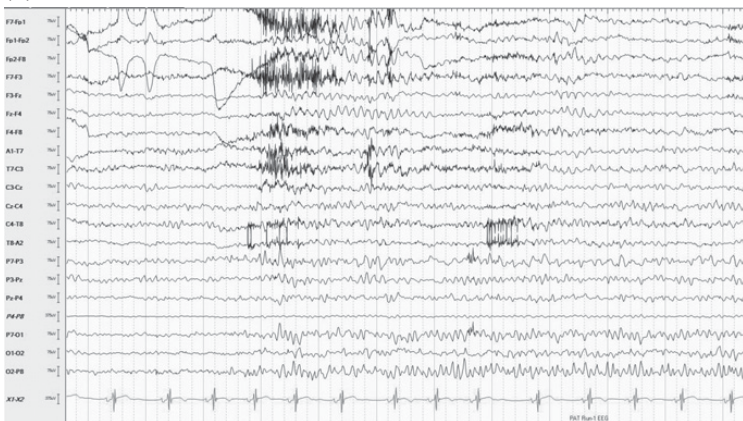


Figure 3.1 (A) Twenty-nine-year-old man with epilepsy, EEG display in longitudinal bipolar montage. This is also the default montage used to display EEGs in this book. (B) Diagram of electrode pairs used in the longitudinal bipolar montage.

Each montage has its distinct strength and weakness. Waveforms of interest should be viewed using both types of montages for accurate localization. Digital displays allow the reader to easily switch between different montages. At the minimum, every EEG must have a longitudinal and transverse bipolar as well as a referential montage [1].

(A)



(B)

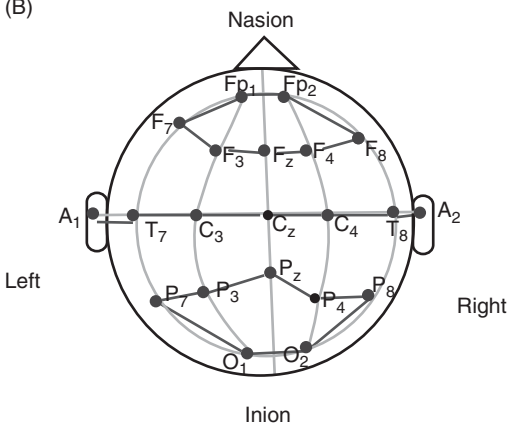
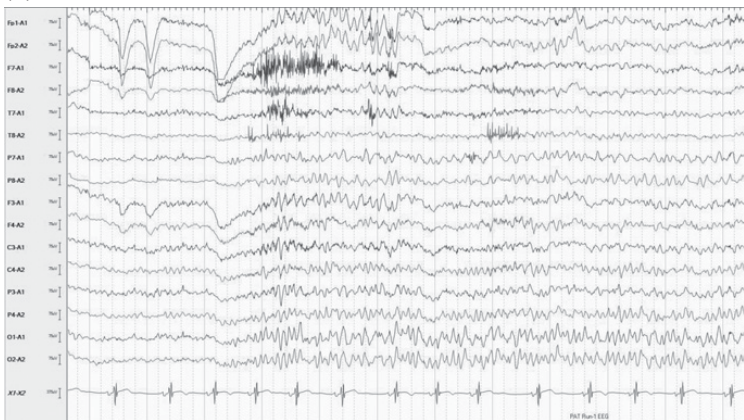


Figure 3.2 (A) Same patient as in Figure 3.1A, EEG display in transverse bipolar montage. (B) Diagram of electrode pairs used in the transverse bipolar montage.

Strengths and Weaknesses

Bipolar montages: As the electrodes in a bipolar montage are adjacent to each other, these montages offer the best “resolution” for focal potentials such as discharges. Another advantage is that they are usually less prone to artifactual

(A)



(B)

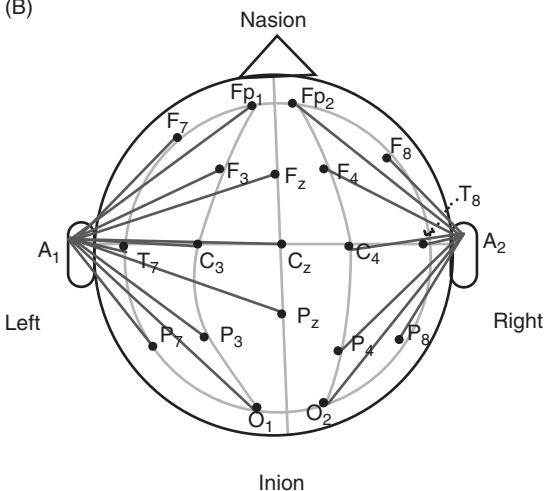
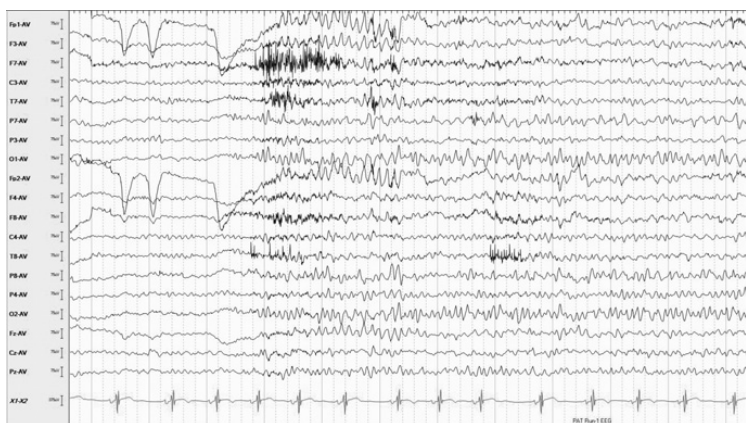


Figure 3.3 (A) Same patient as in Figure 3.1A, EEG display in an ipsilateral ear reference montage. (B) Diagram of electrode pairs used in the ipsilateral ear reference montage.

contamination. This is because artifactual potentials tend to involve adjacent electrodes to a similar degree and thus neutralize each other. On the flip side, their main disadvantage pertains to poor resolution for broad or generalized



potentials. Since these are thinly spread out over a large area, there is only a small potential difference between adjacent electrodes. Therefore, broad or generalized potentials are much harder to appreciate on a bipolar montage.

Referential montage: As the electrodes are far apart, these montages offer the best “resolution” for broad or generalized potentials. Small potential differences are amplified with longer interelectrode distances. On the flip side, referential montages are prone to artifactual contamination as a contaminated reference leads to artifact in all the channels [2]. Understanding montages is the key to successful localization.

Chapter Summary

1. Montages are specific arrangements of channels; a channel is a pair of electrodes.
2. Waveforms result from cerebral potentials; each montage allows the reader to appreciate the same waveform through a distinct point of view.
3. Bipolar montages consist of channels with adjacent electrode pairs.
4. Referential montages consist of channels with nonadjacent electrode pairs. One electrode is located closer to the cerebral activity (active), while the other is intended as a distant reference (inactive).
5. Common bipolar montages include the longitudinal bipolar (double banana) and transverse bipolar.

6. Common referential montages include contralateral and ipsilateral ear reference, central reference (vertex), or average reference.
7. Bipolar montages are less prone to artifact and best appreciate focal potentials.
8. Referential montages are more prone to artifact and best appreciate broad or generalized potentials.
9. An active reference results when the reference which is intended to be neutral is very active.
10. The reader must use both types of montages to evaluate the same waveform. Digital displays make it easy to switch between montages.

References

1. Sinha SR, Sullivan LR, Sabau D, et al. American clinical neurophysiology society guideline 1: minimum technical requirements for performing clinical electroencephalography. *The Neurodiagnostic Journal*. 2016 Oct 1;56(4):235–44.
2. Britton JW, Frey LC, Hopp JL, et al. *Electroencephalography (EEG): an introductory text and atlas of normal and abnormal findings in adults, children, and infants*. American Epilepsy Society, Chicago; 2016.